

$$u_{n+1} = \frac{u_n^2 + u_n + 12}{2u_n}.$$

- (a) By considering $u_{n+1} - 4$, or otherwise, prove by mathematical induction that $u_n > 4$ for all positive integers n . [5]

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(b) Show that $u_{n+1} < u_n$ for $n \geq 1$. [3]

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